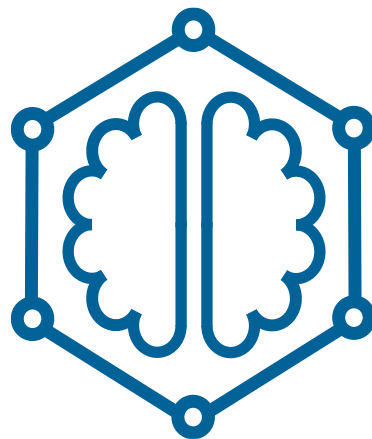
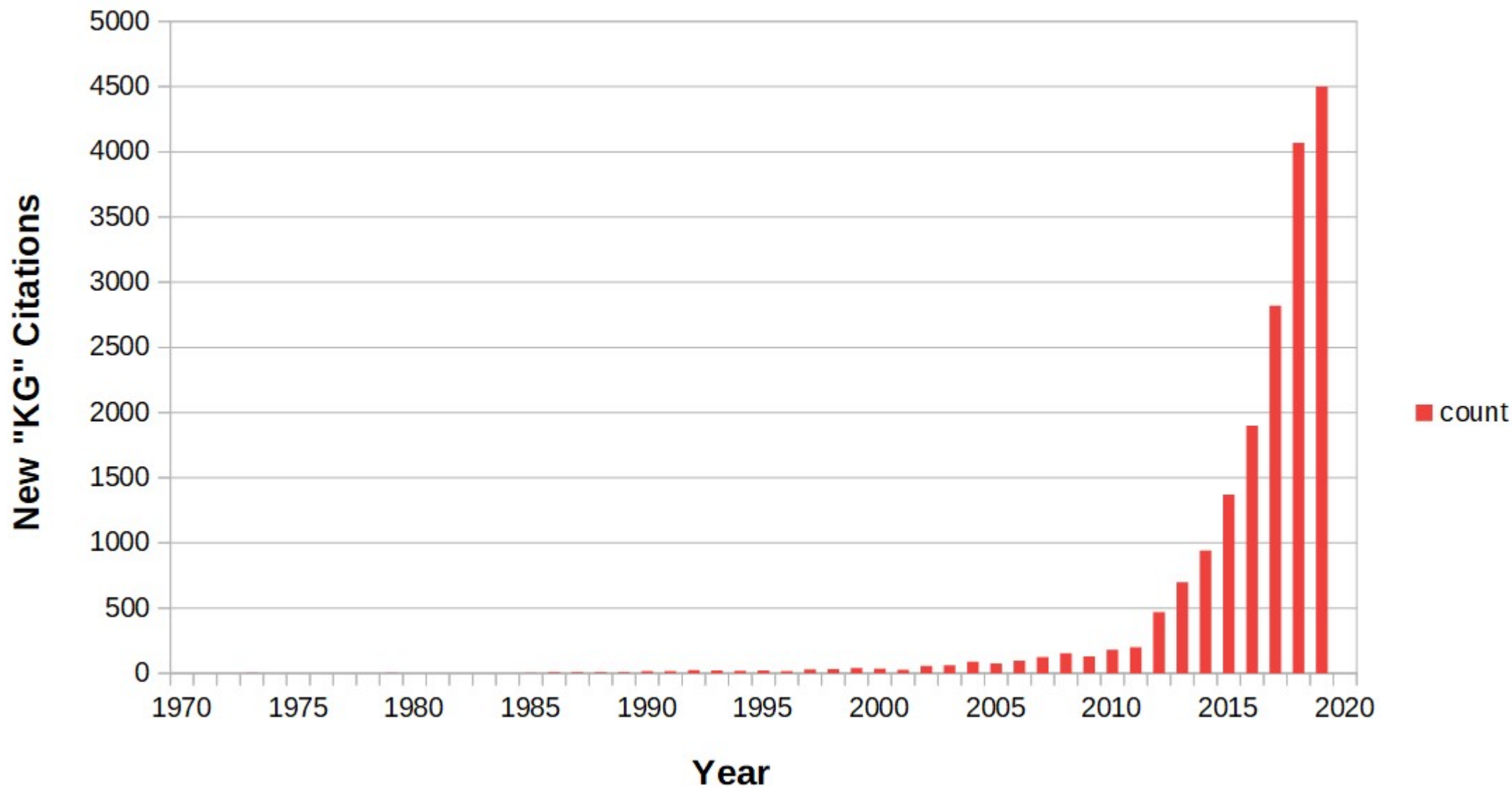




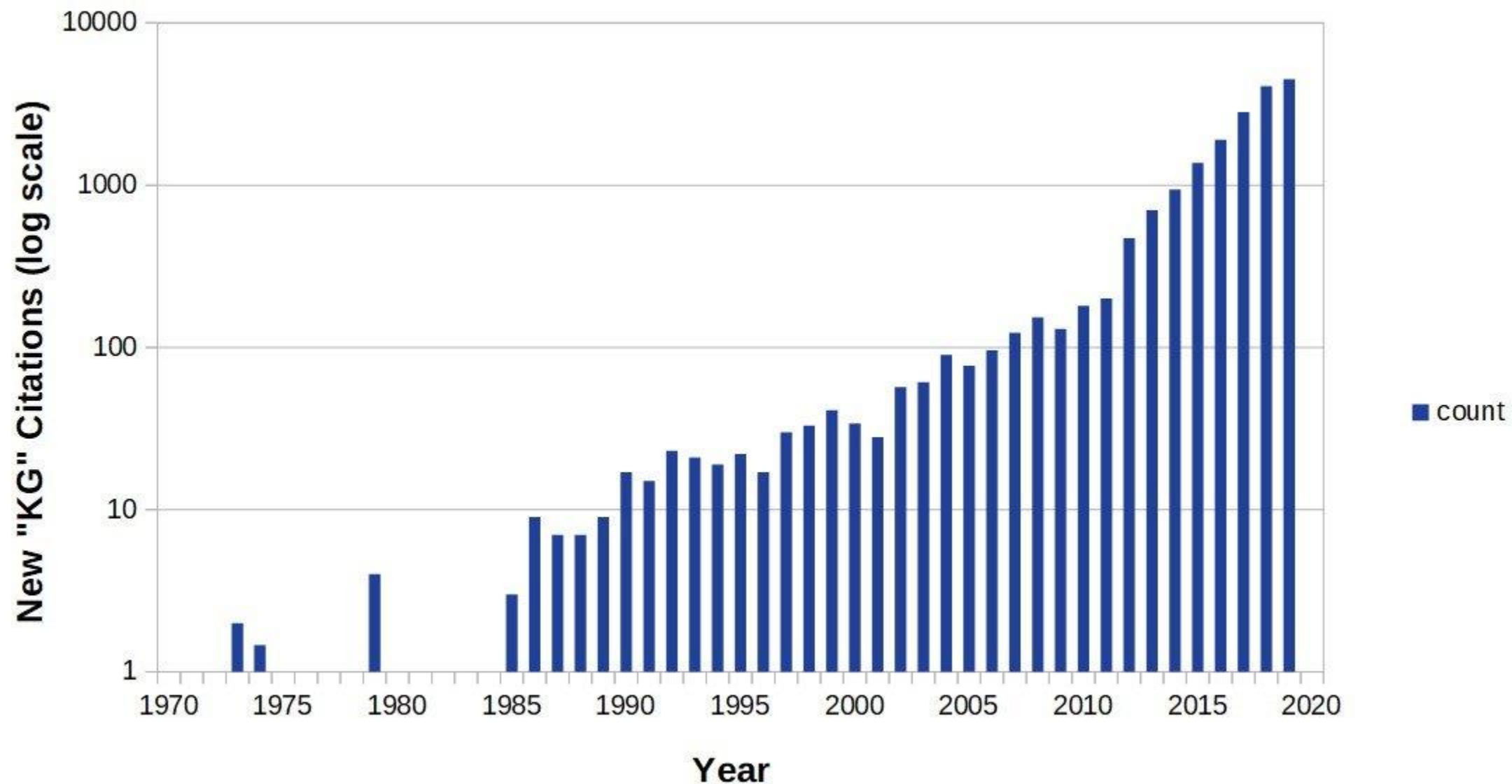
What is a Knowledge Graph?

Understanding the Space

Emanuel Sallinger



Source: <https://www.mkbergman.com/2244/a-common-sense-view-of-knowledge-graphs/>



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Just Graphs?





Is a Graph a Knowledge Graph?

$$G = (V, E)$$



Google's Viewpoint

“A graph that understands real-world entities and
their relationships to one another:
things, not strings”



““Introduction the Knowledge Graph: Things, Not Strings”. A. Singhal. Google Blog 2012.



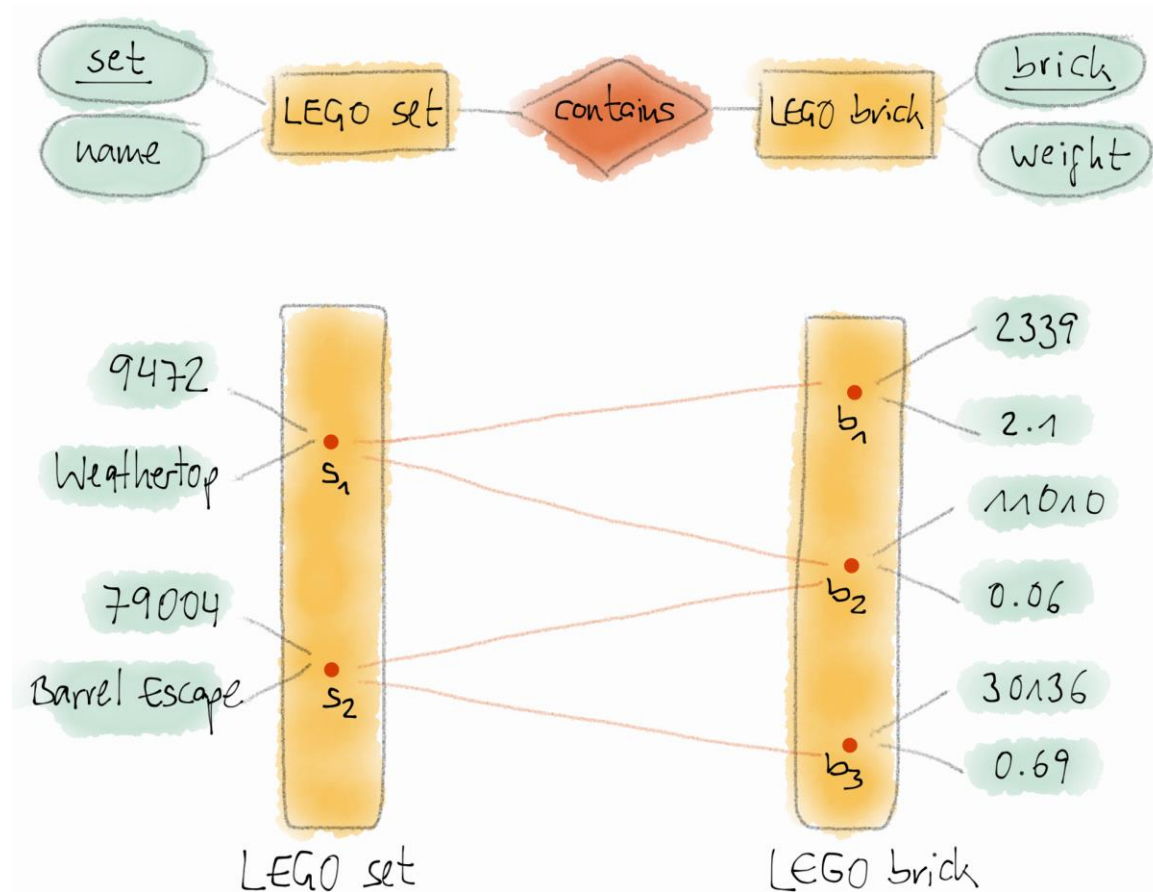
Going Back to 1974

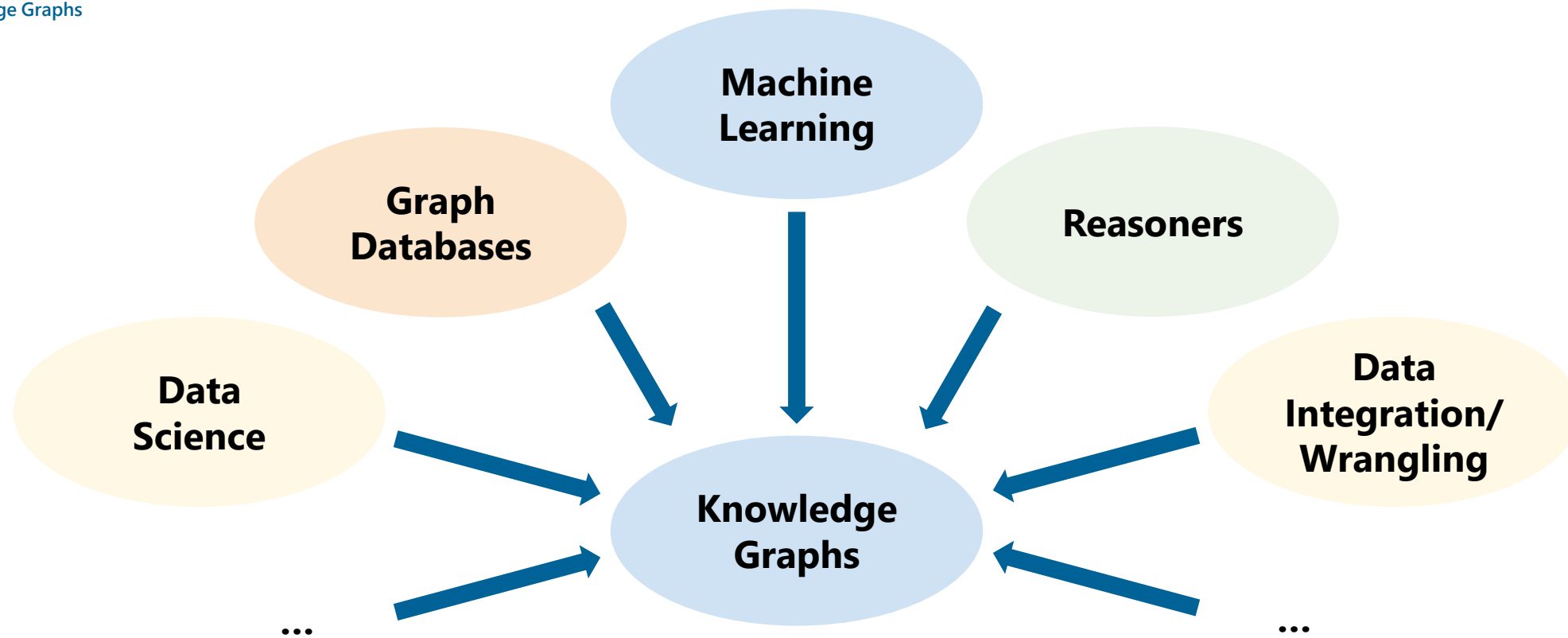
“A mathematical structure with **vertices** as *knowledge units* connected by **edges** that represent the *prerequisite relation*”

“On the Structure of the Teaching-learning Interactive Process”. E. Marchi and O. Miguel. I.J. Game Theory 1974.



Is an ER-Diagram a Knowledge Graph?







The Role of Knowledge





Graph plus Intent

“A **graph** of data with the intent
to compose **knowledge**”

“(Re)Defining Knowledge Graphs”. A. Hogan et al. Dagstuhl 2019.



Distinguishing Properties

“Are characterized by several properties that together *distinguish* them from more traditional knowledge management paradigms:

1. **Normalization:** Information is *decomposed* into small units of information, interpreted as edges of some form of graph.
2. **Connectivity:** Knowledge is represented by the relationships between these units.
3. **Context:** Data is *enriched* with contextual information to record aspects such as temporal validity, provenance, trustworthiness, or other side conditions and details”

“Ontologies for Knowledge Graphs?“, M. Krötzsch. DL 2017.



Providing Support

“An organized and curated set of facts
that **provide support** for
models to understand the world”

“Knowledge Graph Conference”. Columbia University. 2019.



Facebook KG Social graph with people, places, things



Amazon PG knowledge graph of all products



Factual Businesses & places



Wolfram KB World facts + mathematics



RIT People, skills, recruiting



Commercial Banks Customers, Companies, Risks, ...



Rating Agencies Companies, Evaluations, Risks, ...



Applying Knowledge





Components of a KG

“A semi-structured data model characterized by three components:

1. a ground **extensional** component, that is, a set of relational constructs for schema and data (which can be effectively modeled as graphs or generalizations thereof)
2. an **intensional** component, that is, a set of inference rules over the constructs of the ground extensional component
3. a **derived** extensional component that can be produced as the result of the application of the inference rules over the ground extensional component (with the so-called “reasoning” process)”

“Knowledge Graphs and Enterprise AI: the Promise of an Enabling Technology”. L. Bellomarini et al. ICDE 2019.



A Broad Definition

“A **knowledge graph** acquires and integrates information into an **ontology** and applies a **reasoner** to derive new knowledge”

“Towards a Definition of Knowledge Graphs”. L. Ehrlinger and W. Wöß. SEMANTICS 2016.

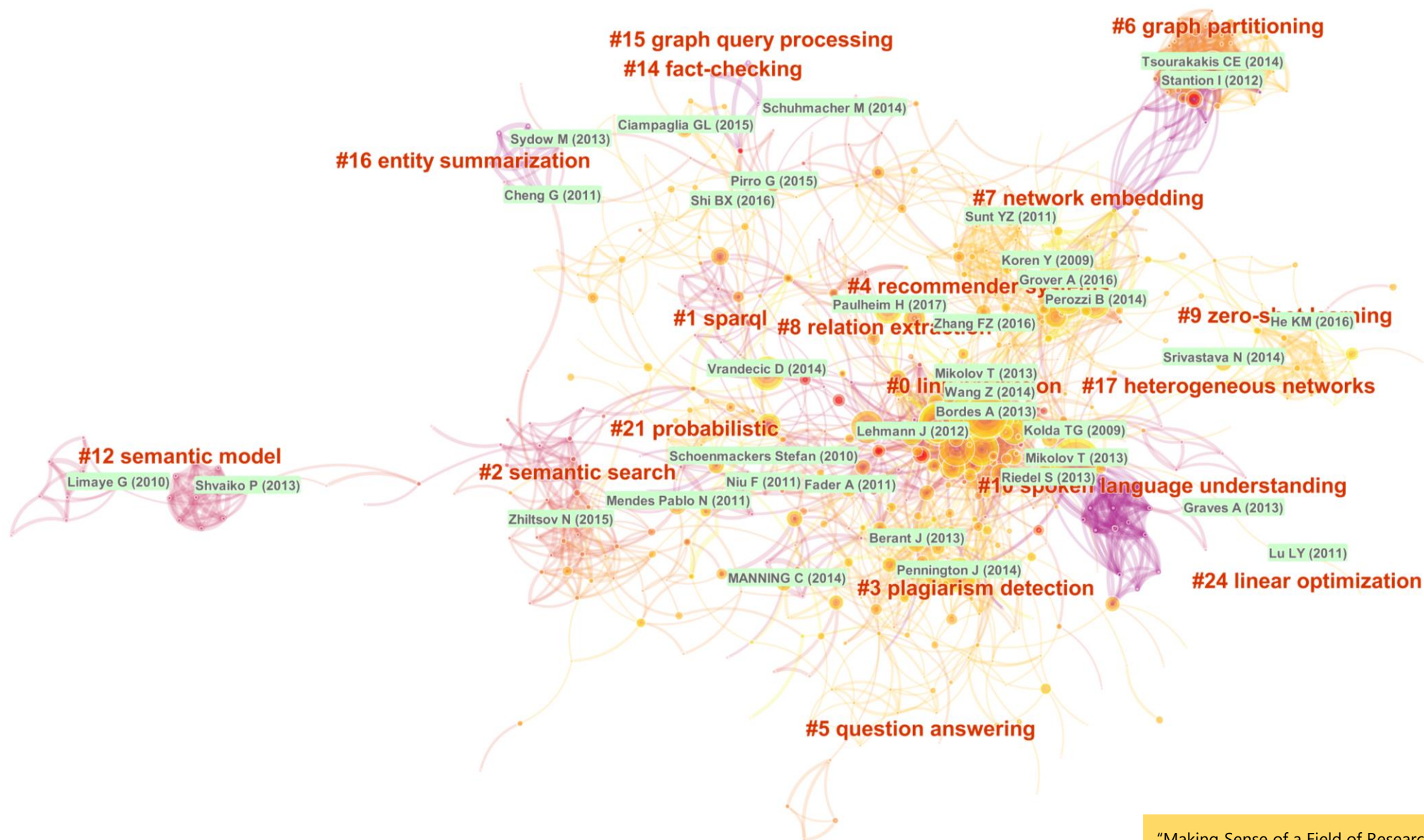


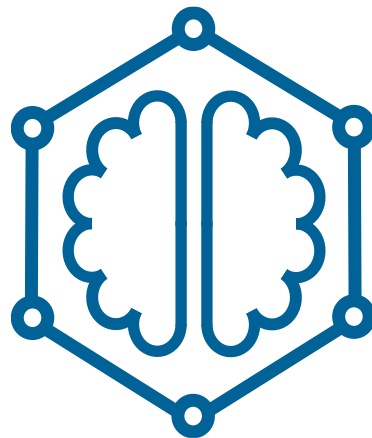
A Broad Definition

“A **knowledge graph** acquires and integrates information into an **ontology** and applies a **reasoner** to derive new knowledge”

scalable

Definition of Knowledge Graphs”. L. Ehrlinger and W. Wöß. SEMANTICS 2016.





What is a Knowledge Graph?

Understanding the Space

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